

Customer Churn Prediction

Machine Learning Report — Corrected Model

1. Objective

Predict whether a telecom customer is likely to churn and use the prediction to prioritize retention actions.

2. Target Variable

ChurnFlag: 0 = Stayed, 1 = Churned.

3. Target Leakage Correction

The initial model produced 100% for all metrics because the encoded feature **Churn_Yes** revealed the answer being predicted. This is target leakage. The corrected workflow removes the original **Churn** column before encoding, then recreates X, y, X_encoded, the train/test split and the model.

4. Data Preparation

customerID is removed as an identifier. Churn is removed to prevent leakage. TenureGroup and MonthlyChargesGroup are excluded because the original numerical fields are available. Missing rows are removed and categorical fields are one-hot encoded.

5. Model

Item	Value
Algorithm	Logistic Regression
Train/Test	80% / 20%
Random State	42
Training Customers	5,634
Testing Customers	1,409

6. Performance

Metric	Result
Accuracy	80.48%
Precision	65.52%
Recall	55.88%
F1 Score	60.32%

7. Confusion Matrix

	Predicted Stay	Predicted Churn
Actual Stay	925 (TN)	110 (FP)
Actual Churn	165 (FN)	209 (TP)

The model correctly identifies 209 churners and 925 retained customers. It misses 165 actual churners and incorrectly flags 110 retained customers.

8. Metric Interpretation

Accuracy: overall correct predictions. **Precision:** how many predicted churners actually churn. **Recall:** how many actual churners were identified. **F1:** balance between precision and recall.

9. Risk Classification

Predicted churn probabilities are converted into business risk levels: High Risk $\geq 70\%$, Medium Risk $40\% - < 70\%$, Low Risk $< 40\%$. These thresholds are project business rules and can be tuned after validation.

10. Business Use

The prediction output can prioritize retention outreach. High-risk customers can receive proactive support, plan reviews, targeted offers or service interventions. The model should support business judgment rather than replace it.

11. Limitations

The model is trained on historical data and probabilities are not guarantees. Performance should be monitored on future data, and retention campaigns should be evaluated to confirm whether model-driven actions reduce churn.

12. Conclusion

After removing target leakage, the Logistic Regression model achieved 80.48% accuracy, 65.52% precision, 55.88% recall and 60.32% F1 score. This provides a realistic ML component for the Customer Churn Business Analyst project.